

# Sinhala Stem/Rest Segmentation

Split Logic and Scoring Heuristics

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## Purpose

The Sinhala Stem Segmenter is designed for modern written Sinhala. It is not a lemmatizer, and it does not delete suffixes. Instead, it segments a word into a left-side *stem* and a right-side *rest*:

stem + rest = normalized word

For example:

ජනාධිපතිවරයා + ය = ජනාධිපතිවරයාය

The goal is to preserve the original word while exposing a useful stem/rest structure for search, tokenization experiments, corpus analysis, and downstream Sinhala NLP tools.

## What Happens When A Sinhala Word Is Fed?

When a word such as ජනාධිපතිවරයාය is given to the segmenter, the model does not simply look up a final answer from a dictionary. It uses stored learned evidence, but the split itself is generated and scored on the fly.

1. **Normalize the word.** The input is Unicode-normalized so equivalent written forms are handled consistently.
2. **Split into Sinhala orthographic clusters.** The word is not split by raw Unicode characters. The segmenter respects the Sinhala abugida writing system and avoids cutting inside dependent vowel signs, al-lakuna, ZWJ/ZWNJ sequences, and related written units.
3. **Generate legal split candidates on the fly.** For ජනාධිපතිවරයාය, the segmenter considers legal boundaries such as:

ජනාධිපති + වරයාය

ජනාධිපතිවරයා + ය

and other legal cluster-safe splits.

4. **Check learned evidence.** The model contains learned tables such as:
  - stem candidate → possible rest parts and frequencies

- rest part → global frequency
  - word → frequency
5. **Score each candidate split.** Each legal split is scored using corpus evidence.
  6. **Choose the best split.** The highest-scoring valid split is selected if it is above the minimum score threshold.
  7. **Fallback if no split is strong enough.** If the model does not find a strong split, it returns the whole word as the stem and an empty rest.

## Example Output

For the input word ජනාධිපතිවරයාය, the model returns:

| Field                 | Value         |
|-----------------------|---------------|
| <b>word</b>           | ජනාධිපතිවරයාය |
| <b>normalized</b>     | ජනාධිපතිවරයාය |
| <b>stem</b>           | ජනාධිපතිවරයා  |
| <b>rest</b>           | ය             |
| <b>boundary</b>       | 12            |
| <b>stem_clusters</b>  | 8             |
| <b>rest_clusters</b>  | 1             |
| <b>confidence</b>     | 1.0           |
| <b>score</b>          | 42.895175     |
| <b>family_size</b>    | 76            |
| <b>family_tokens</b>  | 22839         |
| <b>rest_frequency</b> | 7212294       |
| <b>split</b>          | True          |

The important user-facing result is:

ජනාධිපතිවරයාය → ජනාධිපතිවරයා + ය

## Meaning Of The Advanced Fields

**word** Original input word.

**normalized** Unicode-normalized word used internally.

**stem** Left-side segment selected by the model.

**rest** Right-side remaining segment.

**boundary** Character offset where the stem ends and the rest starts.

**stem\_clusters** Number of Sinhala orthographic clusters in the stem.

**rest\_clusters** Number of Sinhala orthographic clusters in the rest.

**confidence** Model confidence between 0 and 1.

**score** Raw internal evidence score for the chosen split.

**family\_size** Number of different rest parts seen with the same stem.

**family\_tokens** Total frequency supporting this stem family.

**rest\_frequency** Frequency of this rest pattern in the model.

**split** True if a split was selected; False if the whole word was kept as the stem.

## Scoring Formula

For each legal candidate split, the model calculates:

$$\begin{aligned} \text{score} = & 2.2 \cdot \log(1 + \text{family\_size}) \\ & + 0.65 \cdot \log(1 + \text{family\_tokens}) \\ & + 1.15 \cdot \log(1 + \text{rest\_frequency}) \\ & + 0.7 \cdot \log(1 + \text{stem\_as\_word\_frequency}) \\ & + 0.12 \cdot \text{stem\_clusters} \\ & - 0.45 \cdot \max(0, \text{rest\_clusters} - 4) \\ & - \text{tiny\_stem\_penalty} \end{aligned}$$

where:

$$\text{tiny\_stem\_penalty} = \begin{cases} 0.7, & \text{if } \text{stem\_clusters} \leq \text{min\_stem\_clusters} \\ 0, & \text{otherwise} \end{cases}$$

## Logic Behind The Heuristics

The formula is a tunable engineering heuristic. It is not copied from a single published Sinhala stemming paper. However, the signals used in it are motivated by established ideas in unsupervised morphology and stemming.

### Family Size

$$2.2 \cdot \log(1 + \text{family\_size})$$

This is the strongest signal. If a left-side string appears with many different right-side parts, it is likely to be a stem-like unit.

Example pattern:

ආණ්ඩුව+ට, ආණ්ඩුව+ේ, ආණ්ඩුව+ක්

This idea is related to *successor variety*: a high variety of possible continuations after a segment can indicate a boundary.

### Family Tokens

$$0.65 \cdot \log(1 + \text{family\_tokens})$$

This rewards stem families that occur often. It is weighted lower than family size because raw frequency alone can over-favor very common accidental strings.

## Rest Frequency

$$1.15 \cdot \log(1 + \text{rest\_frequency})$$

This rewards common right-side patterns. If a rest part appears across many words, it is a useful segmentation clue.

## Stem As Word Frequency

$$0.7 \cdot \log(1 + \text{stem\_as\_word\_frequency})$$

This rewards stems that also appear independently as words. This is only a soft clue; a useful stem does not have to be a standalone dictionary word.

## Stem Length Reward

$$0.12 \cdot \text{stem\_clusters}$$

This gives a small reward to longer stems, helping avoid splits that happen too early in the word.

## Long Rest Penalty

$$-0.45 \cdot \max(0, \text{rest\_clusters} - 4)$$

This penalizes unusually long rest parts. A very long rest can indicate that the split happened too early.

## Tiny Stem Penalty

Very short stems are penalized because they are more likely to be accidental prefix-like fragments rather than useful stems.

## Why Use Logarithms?

The model uses  $\log(1 + x)$  for count-based signals. This gives diminishing returns:

- frequency evidence helps;
- very large counts do not completely dominate the decision;
- multiple kinds of evidence can still influence the split.

## Literature Support

The exact numeric weights are engineering choices and should be tuned with evaluation data. The underlying ideas, however, have support in computational morphology:

- **Successor variety:** Harris-style segmentation uses the idea that many possible continuations after a segment can indicate a morpheme boundary. This supports the use of *family\_size*.
- **Unsupervised morphology:** Morfessor-style models show that useful morph-like units can be induced from raw text using frequency and form/usage evidence.
- **Conservative stemming:** Stemming for information retrieval often prefers avoiding harmful over-stemming. This motivates penalties for tiny stems and very long rest parts.

## Practical Interpretation

For most users, the core output is:

word → stem + rest

The score and frequency fields are best treated as diagnostic evidence. They help explain why the model selected a split, but they are not grammatical labels and not proof that the split is the only linguistically correct analysis.

## References

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- Creutz, M., & Lagus, K. (2007). *Unsupervised models for morpheme segmentation and morphology learning*. *ACM Transactions on Speech and Language Processing*, 4(1).
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